Justin Lee

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Education

Stony Brook University | **GPA: 3.50/4.0** — Stony Brook, NY
M.S. in Applied Mathematics and Statistics — May 2028
B.S. in Computer Science and Applied Mathematics

Coursework: Data Structures, Advanced Algorithms, Operating Systems, Computer Networks, Artificial Intelligence, Probability Theory, Statistical Learning, Fundamentals of LLMs, Software Engineering

Experience

Betterment — June 2026 – August 2026
Software Engineer Intern — *New York City, NY*

- Shipped payroll lifecycle tooling, **tracing** failed events across **350+** provider integrations and sync workflows.
- Implemented **idempotency** for queued payroll retries, reducing duplicate retry events by **70%** during provider failures.
- Owned 401(k) lifecycle event tracking and Datadog monitoring, **reducing** company-wide on-call triage tickets by **34%**.
- Protected downstream writes with payload validation and **failure isolation** for malformed payroll event records.

Acadia Analytics — May 2025 – August 2025
Software Engineer Intern — *Cambridge, Massachusetts*

- Designed rolling-window pipelines for **30** stocks and ETFs across 20M+ rows spanning 5+ years of market data.
- Cut retraining time **30%** by parallelizing parameter sweeps and pruning configurations after gains stabilized.
- Automated **100+** rolling backtests with first-pass reports, cutting analyst model review time by **25%** overall.
- Optimized REST and SQL access to forecast outputs, reducing p95 dashboard latency **18%** for internal analysts.

Desung — Jan 2024 – August 2024
Software Engineer Intern — *New York City, NY*

- Engineered batch telemetry for **1M+** Sokoban moves, capturing coordinates, player actions, and session outcomes.
- Designed loop detection to **halt** stuck RL agents and extract repeated move sequences with **85%** overall accuracy.
- Incorporated failed trajectories into RL training, improving agent solve rates **20%** across unseen game levels.

Projects

Seawolf Course Finder | *AWS (EC2, RDS), PyTorch, scikit-learn, BeautifulSoup4*

- Launched AWS course search for SBU classes, serving **50K+** monthly student queries at sub-second latency.
- Scraped and normalized catalog and prerequisite data, indexing **4,000+** courses for semantic search and filters.
- Deployed Flask services on **EC2/RDS** with caching to keep search sub-second during registration.

Hive - Multi-Agent Chat | *FastAPI, WebSockets, PostgreSQL, LangChain*

- Orchestrated **5+** LLM agents through configurable parallel and sequential discussion workflows for each chat.
- Integrated **persistent** chat memory for 50+ conversations while **preserving** context across 30+ message threads.
- Reduced token usage by **caching** repeated chat context and reusing persistent memory across agent conversations.

Awards and Leadership

Top 20 Individual Scorer - New York State Mathematics League (NYSML)
Hackathon Winner — Stony Brook University HopperHacks (2025)
Datathon Winner (Financial Analysis Track) — Stony Brook Datathon (2025)
Business Analytics Club — Co-Founder and Secretary
Top 1% Ranked Competitor in Valorant and Fortnite - 6 years of Consistent Standings

Technical Skills

Languages: Python, Java, Ruby, C++, SQL, TypeScript
Backend/Data: Rails, FastAPI, Flask, PostgreSQL, Redis, Node.js, Express
Cloud/Tools: AWS (EC2, RDS, S3), Docker, Linux, PyTorch, scikit-learn, Pandas, LangChain